

Mathematics

(041)



SECTION - A

Q.1:

$$(A) \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}$$

Q.2:

$$(D) [2, \infty)$$

Q.3:

$$(b) \int_0^a f(x) dx$$

Q.4:

$$(D) 8$$

Q.5:

$$(B) \left[-\frac{\pi}{2}, 0 \right]$$

Q.6:

$$(B) \frac{\pi}{3} \text{ only}$$

Q.7:

$$(D) \frac{1 - P(E \cup F)}{P(\bar{E})}$$

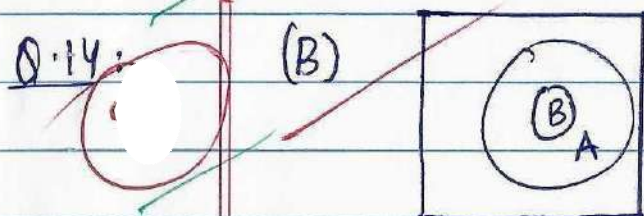
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Question Number

Q.8: (c) Null matrix

Q.9: (c) $x = 3t + 4$, $y = t - 3$, $z = 2t + 7$

Q.10: (B) Bina

Q.11: (c) 1 cm/s.

Q.12: (D) $\frac{2\pi}{3}$ Q.13: (c) $\{1, -5, -6\}$ Q.15: (D) $\int_0^4 \sqrt{y} dy$



Q.16: (C) The objective function maximizes the combined profit earned from selling x & y .

Q.17: (B) $AB = BA$.

Q.18: (B) 1.

Q.19: (D) Assertion (A) is false but Reason (R) is true.

Q.20: (D) Assertion (A) is false but Reason (R) is true.

SECTION - D

Q.32:

Required shaded area = Area of region EAB + Area of region ECD

$$= \int_{-2}^{-2/5} (5x+2) dx + \int_{-2/5}^2 (5x+2) dx$$

$$= \left[\frac{5x^2}{2} + 2x \right]_{-2}^{-2/5} + \left[\frac{5x^2}{2} + 2x \right]_{-2/5}^2$$

$$= \left[\frac{2}{5} + 2\left(-\frac{2}{5}\right) - 10 + 4 \right] + \left[10 + 4 - \frac{2}{5} + \frac{4}{5} \right]$$

$$= \frac{32}{5} + \frac{72}{5}$$

$$= \frac{104}{5} \text{ sq. units.}$$

(Graph on page - 19).



Q.33 :

$$\text{let } I = \int \frac{x^2+x+1}{(x+2)(x^2+1)} dx$$

$$\frac{x^2+x+1}{(x+2)(x^2+1)} = \frac{(x^2+1)}{(x+2)(x^2+1)} + \frac{x}{x^2+1}$$

$$\text{let } I = \frac{x^2+x+1}{(x+2)(x^2+1)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+1}$$

$$x^2+x+1 = A(x^2+1) + (Bx+C)(x+2)$$

$$x^2+x+1 = Ax^2 + A + Bx^2 + 2Bx + Cx + 2C$$

$$x^2+x+1 = (A+B)x^2 + (2B+C)x + (2C+A)$$

On comparing LHS and RHS:-

$$A+B=1 \quad (1) \quad 2B+C=1 \quad (2) \quad 2C+A=1 \quad (3)$$

$$B=1-A$$

Putting $B=1-A$ in (2): $2(1-A)+C=1$

$$2-2A+C=1$$

$$-2A+C=-1 \quad (4)$$



Multiplying ④ with 2 and equating with ③: $-4A + 2C = -2$
 $\frac{2C}{A} + 2C = 1$

$$-5A = -3$$

$$A = \frac{3}{5}$$

$$B = 1 - A = 1 - \frac{3}{5} = \frac{2}{5}$$

$$C = 1 - 2B = 1 - \frac{4}{5} = \frac{1}{5}$$

Putting A, B & C in I:

$$I = \frac{3}{5} \int \frac{dx}{x+2} + \frac{1}{5} \int \frac{2x+1}{x^2+1} dx$$

$$= \frac{3}{5} \log |x+2| + \frac{1}{5} \int \frac{2x}{x^2+1} dx + \frac{1}{5} \int \frac{dx}{x^2+1}$$

$$= \frac{3}{5} \log |x+2| + \frac{1}{5} \log |x^2+1| + \frac{1}{5} \tan^{-1} x + C$$

∴ $\int \frac{x^2+x+1}{(x+2)(x^2+1)} = \frac{3}{5} \log |x+2| + \frac{1}{5} \log |x^2+1| + \frac{1}{5} \tan^{-1} x + C$



Q. 34: (a)

$$\text{let } L_1: \frac{x+1}{2} = \frac{y+1}{1} = \frac{z-9}{-3}$$

$$L_2: \frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$$

Expressing L_1 & L_2 in vector form of equations-

$$\vec{r}_1 \Rightarrow (-\hat{i} + \hat{j} + 9\hat{k}) + \lambda(2\hat{i} + \hat{j} - 3\hat{k})$$

$$\vec{r}_2 = (3\hat{i} - 15\hat{j} + 9\hat{k}) + \mu(2\hat{i} - 7\hat{j} + 5\hat{k})$$

Comparing $\vec{r}_1$ & $\vec{r}_2$ with $\vec{r} = \vec{a} + k\vec{b}$

$$\vec{a}_1 = -\hat{i} + \hat{j} + 9\hat{k}$$

$$\vec{b}_1 = 2\hat{i} + \hat{j} - 3\hat{k}$$

$$\vec{a}_2 = 3\hat{i} - 15\hat{j} + 9\hat{k}$$

$$\vec{b}_2 = 2\hat{i} - 7\hat{j} + 5\hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = 4\hat{i} - 16\hat{j} + 0\hat{k}$$

$$\begin{aligned} \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -3 \\ 2 & -7 & 5 \end{vmatrix} = \hat{i}(5-21) - \hat{j}(10+6) + \hat{k}(-14-2) \\ &= -16\hat{i} - 16\hat{j} - 16\hat{k} \end{aligned}$$



$$\begin{aligned} |\vec{b}_1 \times \vec{b}_2| &= \sqrt{(-16)^2 + (-16)^2 + (-16)^2} \\ &= \sqrt{3(16)^2} \\ &= 16\sqrt{3}. \end{aligned}$$

$$\begin{aligned} (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (4\hat{i} - 16\hat{j} + 0\hat{k}) \cdot (-16\hat{i} - 16\hat{j} - 16\hat{k}) \\ &= -64 + 256 + 0 \\ &= 192. \end{aligned}$$

∴ shortest distance between the given line = $\frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}$

$$= \frac{-192}{16\sqrt{3}}$$

$$= \frac{12\sqrt{3}}{3}$$

$$= 4\sqrt{3} \text{ units.}$$



Q.36: (a)

$$AB = \begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} -4+4+8 & 4-8+4 & -4-8+12 \\ -7+1+6 & 7-2+3 & -7-2+9 \\ 5-3-2 & -5+6-1 & 5+6-3 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

$$= 8I$$

$$\therefore AB = 8I \Rightarrow B^{-1} = \frac{1}{8}A$$

- (1)

The given system of linear equations can be represented in the following manner:-

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$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 9 \\ 1 \end{bmatrix}$$

$$BX = C$$

$$X = B^{-1}C$$

$$= \frac{1}{8} AC$$

(From ①).

$$= \frac{1}{8} \begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 9 \\ 1 \end{bmatrix}$$

$$= \frac{1}{8} \begin{bmatrix} -16 + 36 + 4 \\ -28 + 9 + 3 \\ 20 - 27 - 1 \end{bmatrix}$$

$$= \frac{1}{8} \begin{bmatrix} 24 \\ -16 \\ -8 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix}$$



$$X = \begin{bmatrix} 3 \\ -2 \\ -1 \end{bmatrix} \Rightarrow x=3, y=-2, z=-1.$$

Ans: $x=3, y=-2, z=-1.$

SECTION - E

Q. 36 :

$$S = \{s_1, s_2, s_3, s_4\}, \quad T = \{t_1, t_2, t_3\}$$

No. of elements in $S = n(S) = 4$

No. of elements in $T = n(T) = 3$.

(i) No. of relations from S to $T = 2^{n(S) \times n(T)} = 2^{4 \times 3} = 2^{12}$.

Ans : There can be 2^{12} relations from S to T .

(ii) $f = \{(s_1, t_1), (s_2, t_2), (s_3, t_2), (s_4, t_3)\}$.

The given function f is not one-one / injective as the images of two elements under f i.e. s_2 and s_3 is the same i.e. t_2 , which implies that every element of $\text{domain}(S)$ does not have a distinct image in $\text{codomain}(T)$.

∴ Since f is not injective, it cannot be bijective, even though it is surjective as every element of codomain has a preimage in domain .

Ans : No, f is not bijective as it is not injective.



(iii)



$$R_1 = \{ (s_1, s_2), (s_2, s_4) \}$$

ordered pairs to be added for the function to be reflexive but not symmetric are :

$$(s_1, s_1), (s_2, s_2), (s_3, s_3), (s_4, s_4).$$

R_1 would hence become :

$$R_1 = \{ (s_1, s_1), (s_2, s_2), (s_3, s_3), (s_4, s_4) \}$$

ANSWER

→ (a)

No. of one-one functions from S to $S = \frac{n(S)}{n(S)} P_{n(S)}$

Since $n(S) < n(S)$ [$3 < 4$], no one-one functions are possible from S to S .

Ans: 0 one-one functions are possible from S to S .

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Q. 37

let E_1 , E_2 and E_3 denote the events of Amber, Bonzi and Comet manufacturing the cars. let A denote the event of selecting an electric car manufactured.

$$P(E_1) = 60\% = \frac{6}{10}$$

$$P(A|E_1) = 20\% = \frac{20}{100}$$

$$P(E_2) = 30\% = \frac{3}{10}$$

$$P(A|E_2) = 10\% = \frac{10}{100}$$

$$P(E_3) = 10\% = \frac{1}{10}$$

$$P(A|E_3) = 5\% = \frac{5}{100}$$

(i) (a) Probability a randomly selected car is an electric car
 $= P(A)$

$$= P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3)$$

$$= \frac{6}{10} \cdot \frac{20}{100} + \frac{3}{10} \cdot \frac{10}{100} + \frac{1}{10} \cdot \frac{5}{100}$$

$$= \frac{120 + 30 + 5}{1000}$$

$$= \frac{155}{1000} = 0.155$$



(ii) Probability a randomly selected car ^{which} is electric and ^{is} manufactured by comet = $P(E_3 | A)$

$$= \frac{P(E_3) \cdot P(A|E_3)}{P(E_1) \cdot P(A|E_1) + P(E_2) \cdot P(A|E_2) + P(E_3) \cdot P(A|E_3)}$$

$$= \frac{1 \cdot 5}{10 \cdot 100}$$

$$= \frac{5}{1000}$$

$$= \frac{1}{200}$$

$$= \frac{1}{31} \quad (\text{Ans})$$

(iii) Probability a randomly selected car ^{which} is electric was manufactured by Amber or Bonzi = $1 - P(E_3 | A)$

$$= 1 - \frac{1}{31} = \frac{30}{31} \quad (\text{Ans})$$

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Question Number

Q: 388

$$f(x) = e^x \sin x$$

(i)

$$f'(x) = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x)$$

Put $f'(x) = 0$ for critical points

$$e^x (\sin x + \cos x) = 0$$

Since in $[0, \pi]$, $e^x \neq 0$ for any $x \in [0, \pi]$,

$$\Rightarrow \sin x + \cos x = 0$$

$$\Rightarrow \sin x = -\cos x$$

$$\Rightarrow -\tan x = 1$$

$$\Rightarrow \tan x = -1$$

$$\Rightarrow x = \frac{3\pi}{4}$$

$$f''(x) = e^x (\sin x + \cos x) + e^x (\cos x - \sin x)$$

Interval	Sign of $f'(x)$	Increasing/Decreasing
$[0, \frac{3\pi}{4}]$	+	Increasing
$[\frac{3\pi}{4}, \pi]$	-	Decreasing

Answer: $f(x)$ is increasing in $[0, \frac{3\pi}{4}]$ as $f'(x) > 0$ in this interval and $f(x)$ is decreasing in $[\frac{3\pi}{4}, \pi]$ as $f'(x) < 0$ in this interval.



(ii) let us evaluate $f''(x)$ to check for local maxima / minima / inflection point

$$f'(x) = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x).$$

$$f''(x) = e^x (\sin x + \cos x) + e^x (\cos x - \sin x).$$

$$f''\left(\frac{3\pi}{4}\right) = e^{3\pi/4} \left(\sin \frac{3\pi}{4} + \cos \frac{3\pi}{4} \right) + e^{3\pi/4} \left(\cos \frac{3\pi}{4} - \sin \frac{3\pi}{4} \right)$$

$$= e^{3\pi/4} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) + e^{3\pi/4} \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right)$$

$$= e^{3\pi/4} (-\sqrt{2}).$$

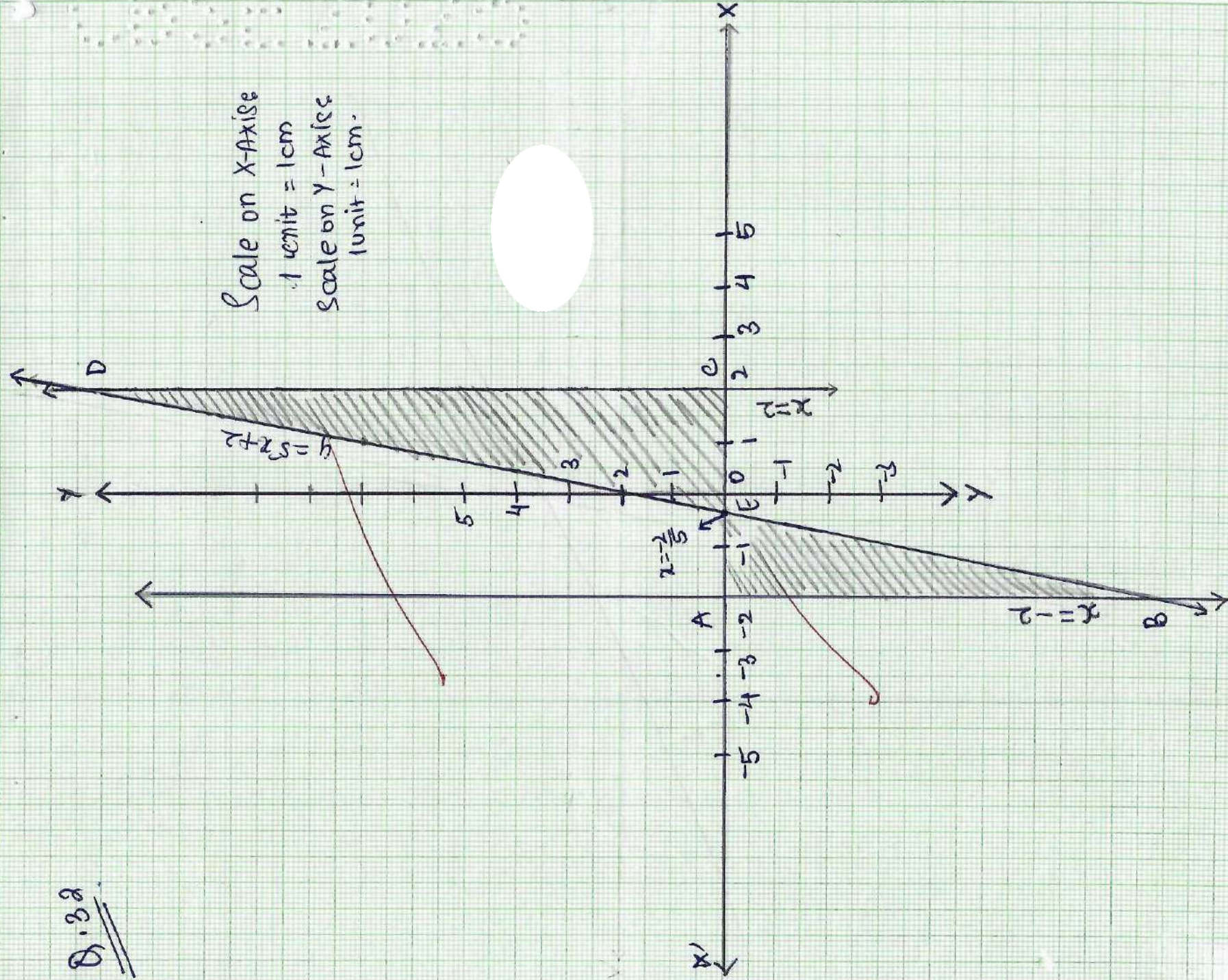
$$= -\sqrt{2} e^{3\pi/4} < 0$$

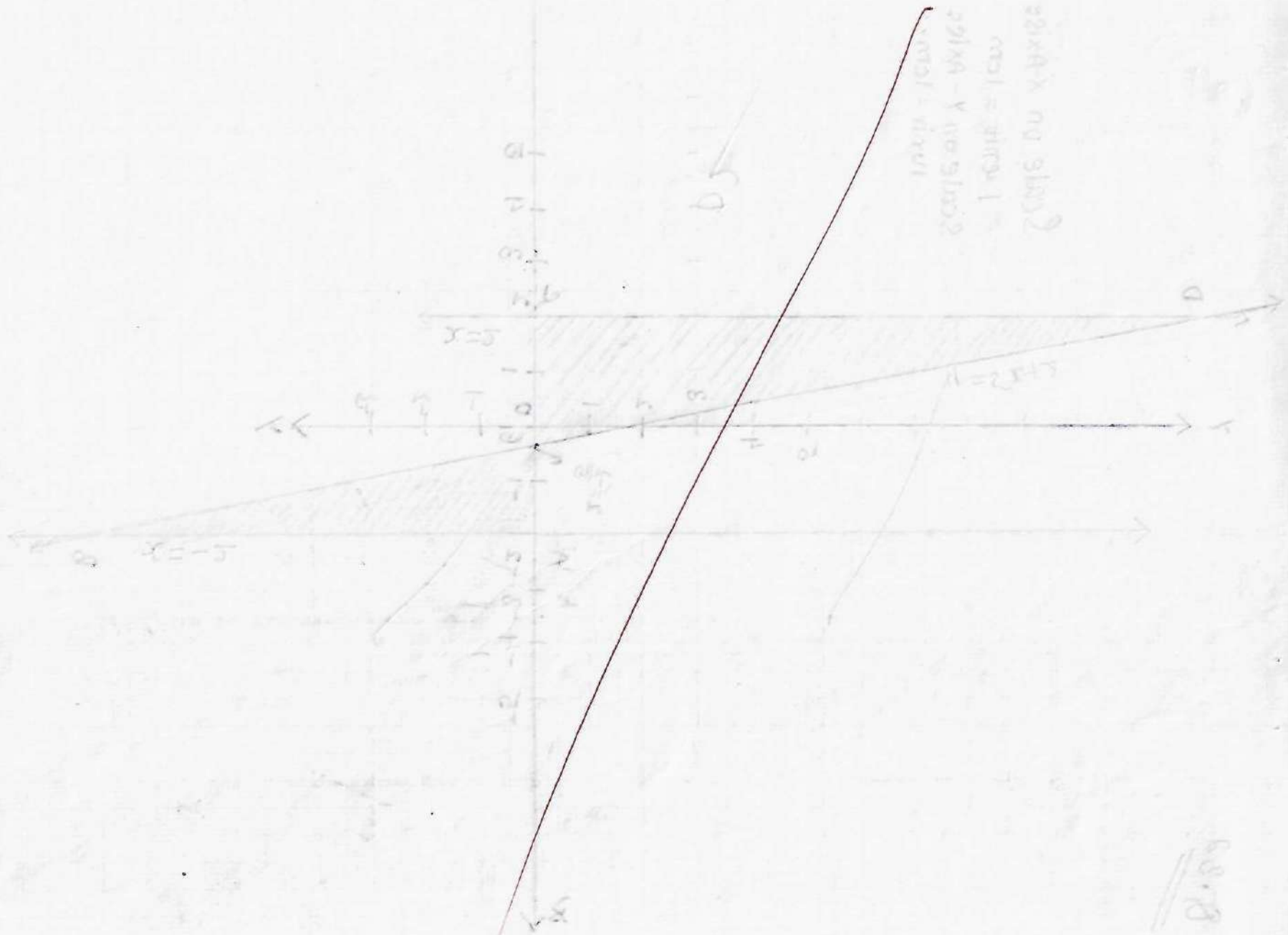
$\Rightarrow x = \frac{3\pi}{4}$ is a point of local maximum.

Answer: As $f''(x) < 0$ at $x = \frac{3\pi}{4}$, it is a point

of local maximum.

Q.32

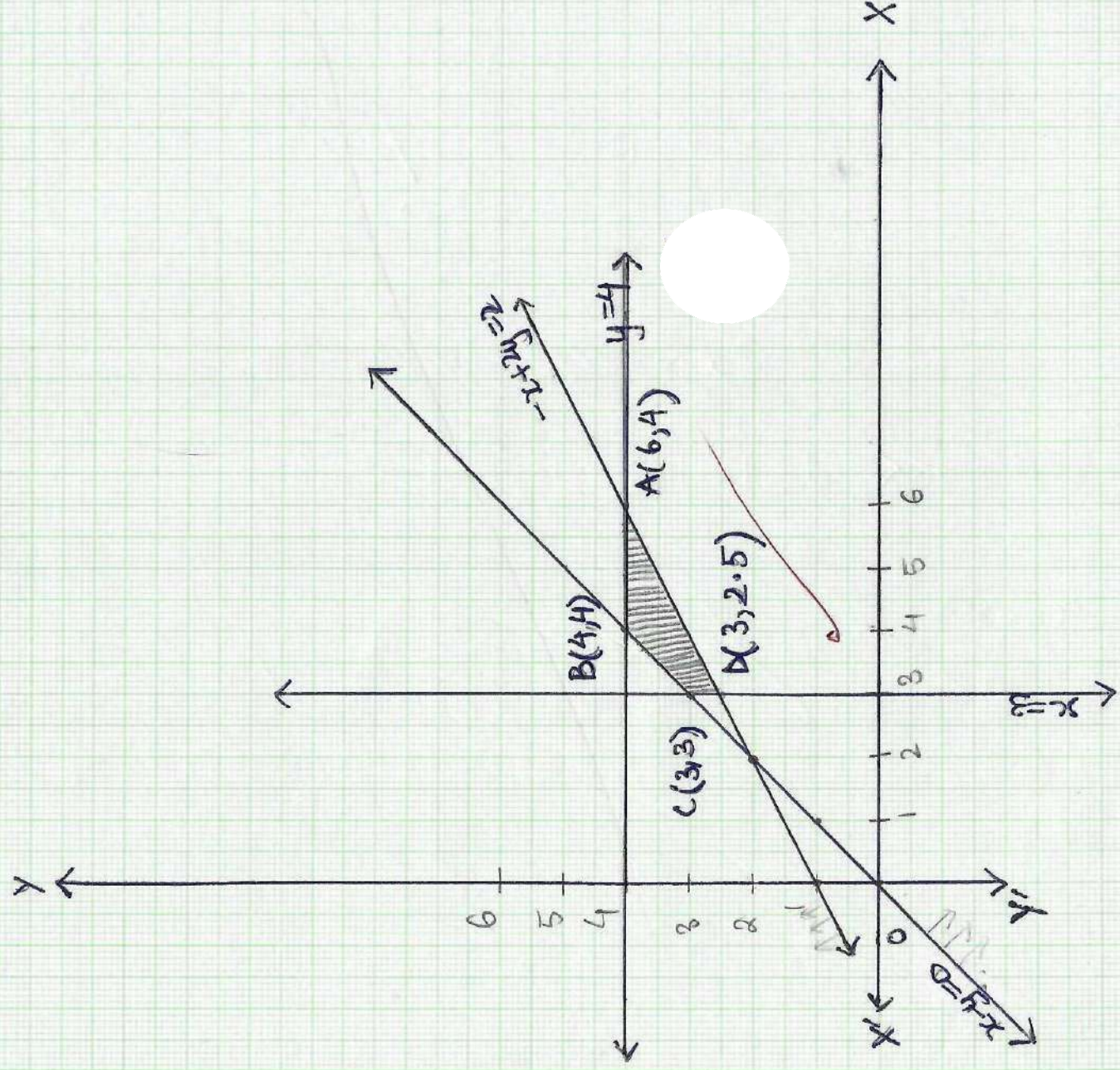




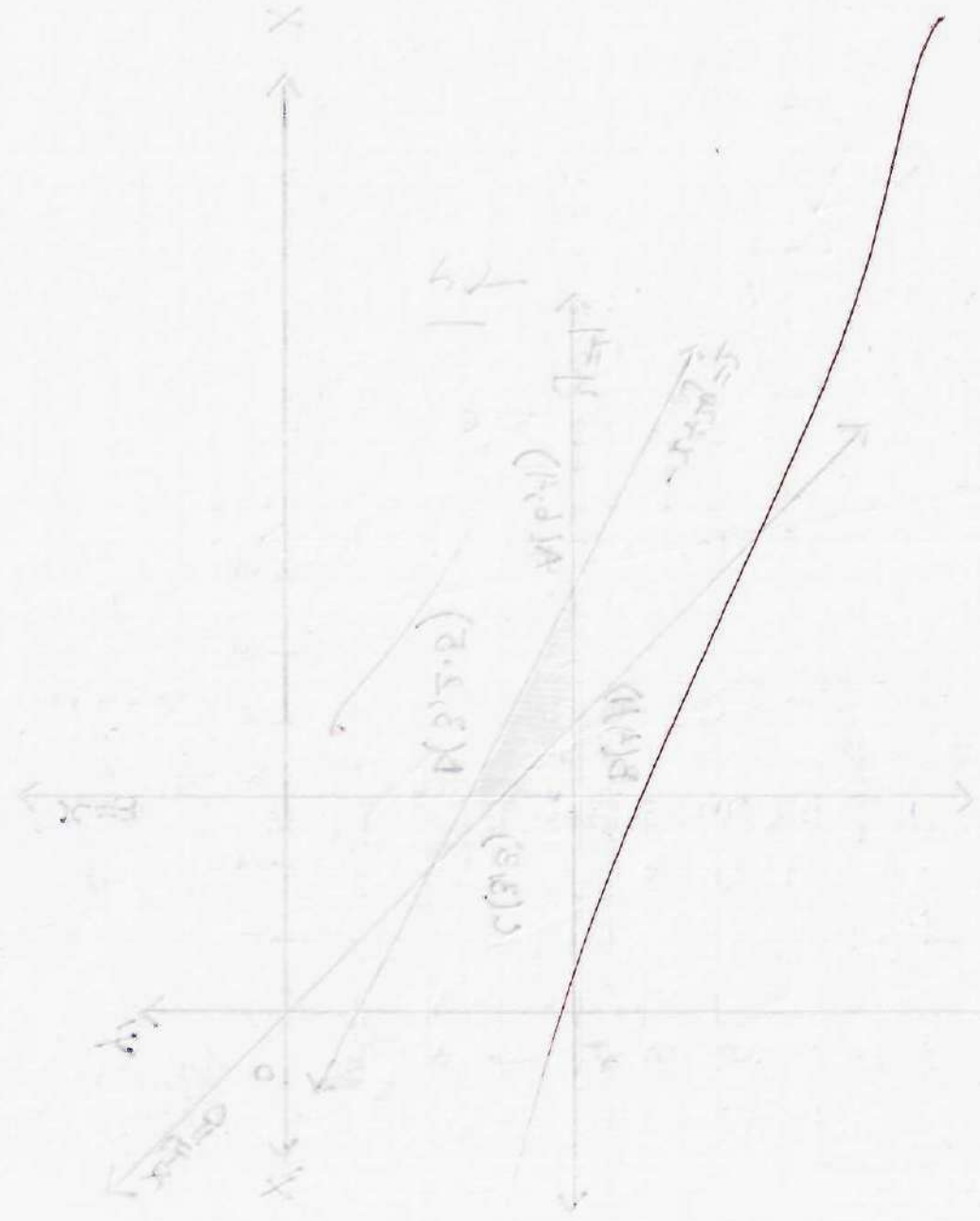
$y = 0$ for $x < -5$
 $y = \frac{1}{10}(x + 5)$ for $-5 \leq x < 5$
 $y = \frac{2}{5}(x - 5) + 1$ for $5 \leq x < 10$

Q.10

Q.28.



EXERCISE



3c.0



SECTION 2

Q. 26: (b) $(1+x^2) \frac{dy}{dx} + 2xy = 4x^2$

$$\frac{dy}{dx} + \left(\frac{2x}{1+x^2} \right) y = \frac{4x^2}{1+x^2}$$

This is a linear differential eqⁿ of the form $\frac{dy}{dx} + Py = Q$, where
 $P = \frac{2x}{1+x^2}$ and $Q = \frac{4x^2}{1+x^2}$.

$$\text{Integrating factor (IF)} = e^{\int P dx} = e^{\int \frac{2x}{1+x^2} dx} = e^{\log|1+x^2|} = (1+x^2).$$

Solution of the differential equation is given by:

$$y(IF) = \int (IF) Q dx + c$$

$$y(1+x^2) = \int (1+x^2) \left(\frac{4x^2}{1+x^2} \right) dx + c$$



$$y(1+x^2) = \int 4x^2 dx + c$$

$$\therefore y(1+x^2) = \frac{4}{3}x^3 + c$$

Ans: The required general solution of the differential equation is $y(1+x^2) = \frac{4}{3}x^3 + c$.

Q.27: $m R n$ if and only if m is a multiple of n , $m, n \in \mathbb{N}$.
 $\Rightarrow R = \{ (m, n) : m = kn, m, n, k \in \mathbb{N} \}$.

Reflexive: Since every natural number is a multiple of itself, $(m, m) \in R$ is true $\forall m \in \mathbb{N}$.

This holds true as $m = km$, where $k=1$.

$\therefore R$ is reflexive.

Symmetric: let $(m, n) \in R$, $m, n \in \mathbb{N}$.

let $m=2$ and $n=1$.

As $m = 2 \times n$, $(2, 1) \in R$.



But $(1, 2) \notin R$ as 1 is not a multiple of 2 and cannot be expressed in the form of $1 = 2k$, where k is a natural number. $\Rightarrow (n, m) \in R$ for $n=1$ and $m=2$.

$\therefore R$ is not symmetric.

(As for all $(m, n) \in R$, $(n, m) \in R$ is not true).

Transitive: let (m, n) and $(n, p) \in R$, $m, n, p \in \mathbb{N}$.

$\Rightarrow$ m is a multiple of n and n is a multiple of p .

$\Rightarrow m = k_1 n$ and $n = k_2 p$, $k_1, k_2 \in \mathbb{N}$.

$\Rightarrow n = \frac{m}{k_1}$ and $n = k_2 p$

$\Rightarrow \frac{m}{k_1} = k_2 p$

$\Rightarrow m = (k_1 k_2) p$, $(k_1 k_2) \in \mathbb{N}$.

$\Rightarrow m$ is a multiple of p .

$\Rightarrow (m, p) \in R$.

$\therefore R$ is transitive.



Q. 288

$$Z = x - 5y \quad (\text{to be minimized}), \quad x, y \geq 0.$$

Constraints:

(i) $x - y \geq 0$

(ii) $-x + 2y \geq 2$

(iii) $x \geq 3$ (iv) $y \leq 4$.

$\Rightarrow x - y = 0$

x	0	1
y	0	1

$\Rightarrow -x + 2y = 2$

x	0	2
y	1	2

The graph of the constraints has been plotted on page 24.
The feasible region is bounded, hence minimum of Z exists and is defined.

Corner Point	$Z = x - 5y$	
A (6, 4)	-14	6 - 20
B (4, 4)	-16 $\rightarrow$ min.	4 - 20
C (3, 3)	-12	
D (3, 2.5)	1	$(\frac{2}{2}) = 1$

Answer: Minimum value of Z is -16 at corner point B(4, 4).

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Question Number

Q.29 : Given: $x\sqrt{1+y} + y\sqrt{1+x} = 0$.

(b) To Prove: $\frac{dy}{dx} = \frac{-1}{(1+x)^2}$

Proof: $x\sqrt{1+y} = -y\sqrt{1+x}$

Squaring both sides

$$x^2(1+y) = y^2(1+x)$$

$$x^2 + x^2y = y^2 + xy^2$$

$$x^2 - y^2 = xy^2 - x^2y$$

$$x^2 - y^2 = xy(y-x)$$

$$(x-y)(x+y) = xy(y-x)$$

$$x+y = -xy$$

$$y+xy = -x$$

$$y(1+x) = -x$$

$$y = \frac{-x}{1+x}$$

PTO →



$$y = \frac{-x}{1+x}$$

Differentiating both sides w.r.t x .

$$\begin{aligned} \frac{dy}{dx} &= \frac{-(1+x) - (-x)}{(1+x)^2} \\ &= \frac{-1-x+x}{(1+x)^2} \\ &= \frac{-1}{(1+x)^2} \end{aligned}$$

Hence, proved.

Q.30 :
(a)

$$P(2) = \frac{3}{10}$$

$$P(\{1, 3, 4, 5, 6\}) = 1 - \frac{3}{10} = \frac{7}{10}$$

$$P(1) = P(3) = P(4) = P(5) = P(6) = \frac{1}{5} \times \frac{7}{10} = \frac{7}{50}$$

let X denote the number of times number 2 appears on the throw of a dice twice. X is a random variable and can take the values 0, 1 or 2.

favourable outcomes when 2 appears once or twice :-

$$\{(1,2), (2,2), (3,2), (4,2), (5,2), (6,2), (2,1), (2,3), (2,4), (2,5), (2,6)\}.$$

n = no. of outcomes when 2 appears atleast once = 11.

Total outcomes on the throw of die twice = 36.

$$P(X=0) = P(\text{not getting a two even once}) = \frac{25 \times \frac{1}{50} \times \frac{1}{50}}{36} = \frac{25}{36}$$

$$P(X=1) = P(\{(1,2), (3,2), (4,2), (5,2), (6,2), (2,1), (2,3), (2,4), (2,5), (2,6)\})$$

$$= \frac{10 \times \frac{1}{50} \times \frac{3}{10}}{36} = \frac{1}{6}$$

$$P(X=2) = P(\{(2,2)\}) = \frac{3}{10} \times \frac{3}{10} = \frac{9}{100}$$

Probability distribution of X :

X	0	1	2
$P(X)$	$\frac{25}{36}$	$\frac{1}{6}$	$\frac{9}{100}$



Q.20 Mean no. of times 2 appears on the dice if thrown twice

$$= E(X) = \sum P(x) \cdot x$$

$$= 0 \cdot \frac{49}{100} + 1 \cdot \frac{42}{100} + 2 \cdot \frac{9}{100}$$

$$= \frac{0 + 42 + 18}{100}$$

$$= \frac{60}{100}$$

$$= 0.6$$

Ans: $E(X) = 0.6$ or $\frac{3}{5}$

Q.21

$$\text{let } I = \int \frac{1}{x} \int \frac{x+a}{x-a} dx$$

$$= \int \frac{1}{x} \int \frac{x+a}{x-a} \cdot \frac{x}{x+a} dx$$

$$= \int \frac{1}{x} \int \frac{(x+a)^2}{x^2-a^2} dx$$



$$= \int \frac{1}{x} \frac{(x+a)}{\sqrt{x^2-a^2}} dx$$

$$= \int \frac{x}{x\sqrt{x^2-a^2}} dx + \int \frac{a}{x\sqrt{x^2-a^2}} dx$$

$$= I_1 + I_2.$$

$$I_1 = \int \frac{x}{x\sqrt{x^2-a^2}} dx = \int \frac{dx}{\sqrt{x^2-a^2}} = \log |x + \sqrt{x^2-a^2}| + c_1.$$

$$I_2 = \int \frac{adx}{x\sqrt{x^2-a^2}}$$

$$\text{let } x = \frac{a}{t} \Rightarrow dx = \frac{-a}{t^2} dt.$$

$$I_2 = \int \frac{a\left(\frac{-a}{t^2}\right) dt}{\frac{a}{t} \sqrt{\frac{a^2}{t^2} - a^2}} = \int \frac{\left(\frac{-a^2}{t^2}\right) dt}{\frac{a}{t} \sqrt{\frac{a^2 - a^2 t^2}{t^2}}}$$

P.T.O. →



$$= \int \left(\frac{a^2}{t^2} \right) dt = - \int \frac{dt}{\sqrt{1-t^2}}$$

$$= - \sin^{-1} t + c_2$$

$$= - \sin^{-1} \left(\frac{a}{x} \right) + c_2$$

$$I = I_1 + I_2$$

$$= \log |x + \sqrt{x^2 - a^2}| + c_1 - \sin^{-1} \left(\frac{a}{x} \right) + c_2$$

$$= \log |x + \sqrt{x^2 - a^2}| - \sin^{-1} \frac{a}{x} + c,$$

$$\text{Where } c = c_1 + c_2.$$

Answer e $\int \frac{1}{x} \sqrt{\frac{x+a}{x-a}} dx = \log |x + \sqrt{x^2 - a^2}| - \sin^{-1} \left(\frac{a}{x} \right) + c$

SECTION-B

Q.21%

(a)

let vector $\vec{a}$ make angles θ with all the three axes.

$$\Rightarrow \cos^2 \theta + \cos^2 \theta + \cos^2 \theta = 1$$

$$\Rightarrow 3 \cos^2 \theta = 1$$

$$\Rightarrow \cos \theta = \pm \frac{1}{\sqrt{3}}$$

Direction cosines of $\vec{a} = \pm \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$.

$$\begin{aligned} \text{Unit vector along the direction of } \vec{a} = \hat{a} &= \pm \left(\frac{1}{\sqrt{3}} \hat{i} + \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k} \right) \\ &= \pm \frac{1}{\sqrt{3}} (\hat{i} + \hat{j} + \hat{k}). \end{aligned}$$

Given: magnitude of vector $\vec{a} = 5\sqrt{3}$.

$$\therefore \vec{a} = |\vec{a}| \cdot \hat{a} = \pm \frac{5\sqrt{3}}{\sqrt{3}} (\hat{i} + \hat{j} + \hat{k}) = \pm 5 (\hat{i} + \hat{j} + \hat{k}).$$

Answer: The required vectors are:

$$\vec{a} = \pm 5 (\hat{i} + \hat{j} + \hat{k}).$$



Q.22

$$\text{let } I = \int_0^{\pi/4} \sqrt{1 + \sin 2x} \, dx$$

$$= \int_0^{\pi/4} \sqrt{\sin^2 x + \cos^2 x + 2 \sin x \cos x} \, dx$$

$$= \int_0^{\pi/4} \sqrt{(\sin x + \cos x)^2} \, dx$$

$$= \int_0^{\pi/4} (\sin x + \cos x) \, dx$$

$$= \left[-\cos x + \sin x \right]_0^{\pi/4}$$

$$= \left[\left(-\cos \frac{\pi}{4} + \sin \frac{\pi}{4} \right) - \left(-\cos 0 + \sin 0 \right) \right]$$

$$= \left[\left(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) - (-1 + 0) \right]$$

$$= 1$$

Answer: $\int_0^{\pi/4} \sqrt{1 + \sin 2x} \, dx = 1$

Q. 23 %

$$f(x) = \sin x - ax + b$$

$$f'(x) = \cos x - a$$

for f to be increasing on $\mathbb{R}$:

$$f'(x) \geq 0$$

$$\cos x - a \geq 0$$

$$a \leq \cos x.$$

Since the maximum value of $\cos x = 1$,

$$a \in (-\infty, 1].$$

$\therefore$ a can take the values from set $(-\infty, 1]$ for
 $f(x)$ to be increasing on $\mathbb{R}$.

Answer: $a \in (-\infty, 1]$



$$\cos x - 0 > 0.$$

$$a \leq \cos x$$

$$\cos x - a \geq 0$$

$$\cos x = a$$

$$\cos x - a \geq 0$$

$$a < \cos x$$

$$1 \geq \cos^{-1} a$$

$$a < 1$$

$$(-1)$$

$$-1, 1$$

$$(-\infty, 1] \quad (-\infty, 1]$$



Q.24 :

$$\vec{a} = (x-2)\vec{a} + \vec{b}$$

$$\vec{b} = (3+2x)\vec{a} - 2\vec{b} \quad \text{PTD} \rightarrow \text{for the answer}$$

$$\text{let } \vec{a} = p_1\hat{i} + q_1\hat{j} + r_1\hat{k}$$

$$\text{and } \vec{b} = p_2\hat{i} + q_2\hat{j} + r_2\hat{k}$$

$$\begin{aligned} \vec{a} &= (x-2)(p_1\hat{i} + q_1\hat{j} + r_1\hat{k}) + p_2\hat{i} + q_2\hat{j} + r_2\hat{k} \\ &= (xp_1 - 2p_1 + p_2)\hat{i} + (xq_1 - 2q_1 + q_2)\hat{j} + (xr_1 - 2r_1 + r_2)\hat{k} \end{aligned}$$

$$\begin{aligned} \vec{b} &= (3+2x)(p_1\hat{i} + q_1\hat{j} + r_1\hat{k}) - 2(p_2\hat{i} + q_2\hat{j} + r_2\hat{k}) \\ &= (3p_1 + 2xp_1 - 2p_2)\hat{i} + (3q_1 + 2xq_1 - 2q_2)\hat{j} + (3r_1 + 2xr_1 - 2r_2)\hat{k} \end{aligned}$$

Since

PTD →

Space for writing
Question Number

Q.24 e

Since $\vec{a}$ & $\vec{b}$ are parallel collinear:

$$\vec{a} = k\vec{b}$$

$$(x-2)\vec{a} + \vec{b} = k[(3+2x)\vec{a} - 2\vec{b}]$$

$$x\vec{a} - 2\vec{a} + \vec{b} = 3k\vec{a} + 2kx\vec{a} - 2k\vec{b}$$

$$(x-2)\vec{a} + \vec{b} = (3k+2kx)\vec{a} - 2k\vec{b}$$

$$\begin{aligned} 2k &= 1 \\ k &= \frac{1}{2} \end{aligned}$$

$$-2k = 1 \Rightarrow k = -\frac{1}{2}$$

$$3k + 2kx = x - 2$$

$$\frac{-3}{2} - x = x - 2$$

$$2x = 2 - \frac{3}{2} = \frac{1}{2}$$

$$x = \frac{1}{4}$$

or

Answer: $x = \frac{1}{4}$



Q.23%

(a)

Given: $e^{x/y} = x$

To prove: $\frac{dy}{dx} = \frac{x-y}{x \log x}$

Proof: $e^{x/y} = x$
 $\log \left(e^{x/y} \right) = \log x$
 $\frac{x}{y} = \log x$

Differentiating both sides w.r.t. x :

$$\frac{1}{y^2} \left(y - x \frac{dy}{dx} \right) = \frac{1}{x}$$

$\Rightarrow$ P.T.O.

$$\frac{x dy}{dx} = y - \frac{1}{x} = \frac{xy - 1}{x}$$

$$\frac{dy}{dx} = \frac{xy - 1}{x^2}$$



$$y - x \frac{dy}{dx} = \frac{y^2}{x}$$

$$x \frac{dy}{dx} = y - \frac{y^2}{x} = \frac{xy - y^2}{x}$$

$$\frac{dy}{dx} = \frac{xy - y^2}{x^2}$$

$$\text{LHS: } \frac{dy}{dx} = \frac{xy - y^2}{x^2}$$

$$\text{RHS: } \frac{x-y}{x \log x} = \frac{x-y}{x \left(\frac{x}{y} \right)} = \frac{xy - y^2}{x^2}$$

∴ LHS = RHS
Hence, proved.



ROUGH

 $P(\bar{E} \cap \bar{F})$

$$(A+B)(A+B)$$

$$A^2 + AB + BA + B^2 = 0$$

$$-4B^2 \quad x=2$$

$$AB - BA = 0 \quad -1 + 1 = 0$$

$$\frac{2-1}{2}$$

$$P(\bar{E})$$

$$P(\bar{E} \cup \bar{F})$$

$$P(\bar{F})$$

$$x=0 \Rightarrow y=5 \Rightarrow z=6$$

$$2\sqrt{1+y} = -y \Rightarrow x$$

$$x^2(1+y) = y^2(1+x)$$

$$bx^2 + x^2y = y^2x$$

$$1, -5, -6 \quad x=4 \Rightarrow y+3 = \frac{z-7}{2} = 1$$

$$y = 2x^3$$

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$$\begin{bmatrix} 1 \\ 6x \\ 8x \end{bmatrix}$$

$$12$$

$$4y$$

$$5$$

$$2x$$

$$4$$

$$4y$$

$$\begin{bmatrix} 1 \\ 12 \\ 4y \end{bmatrix}$$

$$6x$$

$$8x$$

$$4$$

$$6$$

$$36 + 4$$

$$t-3$$

$$2t+7$$

$$2x-y$$

$$2$$

$$4y = 8x$$

$$1 + 1 + \cos 60^\circ = 2$$

$$y = 4$$

$$A+B=1$$

$$B = \frac{1-3}{2} = -1 \Rightarrow V = \frac{1}{2} \Rightarrow \pi r^2 h$$

$$SA = \frac{2}{5} \Rightarrow 100\pi h$$

$$\frac{dv}{dt}$$

$$100\pi \cdot \frac{dh}{dt} = 100\pi$$

$$|p+q|^2 = 1+1+2pq = 1$$

$$2 \cos A = 1$$

$$pq = -\frac{1}{2}$$

$$\cos \theta = \frac{1}{2}$$